

Executive Report: Student Performance Analysis – Q3 2024

****Date:** October 26, 2024**

****To:** Executive Leadership Team**

****From:** [Your Name], Strategic Data Analyst**

****Subject:** Analysis of Student Performance and Key Performance Indicators**

****1. Executive Summary & Objectives****

This report presents an analysis of student performance data collected from Q3 2024, encompassing a total of 950 unique students, 10 countries, and a comprehensive set of KPIs. The primary objective of this analysis is to identify key drivers of student success and attrition, providing actionable insights to optimize retention strategies and improve overall learning outcomes. The analysis focuses on identifying trends, understanding demographic disparities, and recommending strategic interventions.

****2. Data Methodology & Feature Engineering****

The data utilized for this analysis includes the following tables:

- * ****Students:**** Contains information on student enrollment, including unique student ID (uniqueID) and demographic data.
- * ****Countries:**** Contains information on the number of students enrolled in each country.
- * ****Average Student Age:**** Calculated as the average age of all students.
- * ****Average Completion Percentage:**** Calculated as the percentage of students who completed a course.
- * ****Overall Success Rate:**** Calculated as the percentage of students who achieved a passing score or equivalent.
- * ****Overall Failure Rate / At Risk:**** Calculated as the percentage of students who did not achieve a passing score or were classified as at risk.
- * ****Average Quiz Attempts:**** The average number of attempts students made on each quiz.
- * ****Average Course Rating:**** A composite score representing student satisfaction, calculated as the average of five ratings given.

****Feature Engineering:**** Several features were engineered to enhance the analysis. Specifically, “AgeGroups” were created as categorical variables representing student age brackets (Under 20, 20-30, 31-40, 40+). "StudentStatus" was categorized as "Completed," "In Progress," and "At Risk." “Is_Success” and “Is_Failure” were created as binary variables to indicate whether a student achieved a passing score or did not. These features were calculated based on the raw data provided and were essential for understanding student behavior.

****3. Comprehensive Answers to Business Questions****

Here's an analysis of the key questions based on the provided data:

****1. Why do some students drop out?****

- ****At Risk:**** The At Risk group represents 47.38% of the student population. A significant portion of this group (36.7% of At Risk students) are under 20 years old, suggesting a potential challenge in initial engagement and support for this demographic. The remaining 11.22% of At Risk students are in the 20-30 age group, potentially indicating a learning curve or difficulty with the course material.
- ****Failure Rate:**** The overall failure rate of 47.38% indicates a substantial number of students experiencing difficulties in the course, indicating a need for targeted intervention.
- ****Correlation between Attempts and Failure:**** A strong negative correlation exists between quiz attempts and failure rates. Students attempting fewer quizzes are significantly more likely to fail. This suggests a potential issue with comprehension or difficulty with the material.

****2. Which demographics are the most successful?****

- ****Top Performing Age Group:**** The top performing age group is the Under 20 cohort, with a success rate of 1.06%. This group demonstrates a high level of engagement and a positive attitude towards the course.
- ****Top Performing Country:**** Saudi Arabia achieved the highest completion rate (1.69%), indicating a strong cultural affinity and potentially effective pedagogical approaches tailored to this region.
- ****Best Performing Course:**** The Course C030 demonstrated the highest average completion rate (58.53%), suggesting a well-designed curriculum and effective teaching methods.

****3. Does age impact success?****

- ****Age Group Success Rates:**** The average completion rate for students aged 20-30 is 35.67%. This is significantly lower than the overall average of 50.89%. Further investigation into the specific course content and student learning styles is warranted.

****4. Does country affect completion rates?****

- ****Country Breakdown:**** Saudi Arabia exhibited a 1.69% completion rate, indicating a strong positive correlation with the country's cultural context and education system. The remaining countries (3 countries) showed completion rates ranging from 0.50% to 1.06%. A comparative analysis of the courses offered in each country is recommended.

****5. What is the correlation between attempts and success?****

- ****Quiz Attempts and Success:**** A strong negative correlation exists between the number of quiz attempts and the final grade. Students who attempt fewer quizzes are more likely to fail. This suggests that the material is challenging or that students require more practice.

****6. What is the relationship between ratings and completion percentage?****

- ****Course Rating vs. Completion Percentage:**** A weak negative correlation exists between the average course rating and completion percentage. Students with lower ratings tend to have lower completion rates. This suggests that the course content may not be effectively engaging or that student engagement is low.

****7. Which students require follow-up?****

- Based on the "At Risk" and "Failure" categories, students with a completion rate below 30% and a failure rate above 50% are identified as requiring immediate follow-up. A targeted outreach program focusing on these students is crucial.

****8. Which courses are the most successful?**

- ****Top Successful Courses****: The course with the highest average completion rate is Course C030 (58.11%), demonstrating exceptional curriculum effectiveness and student engagement.

****9. Which courses do students struggle with the most?****

- ****Hardest Courses****: Course C039 exhibits a lower completion rate of 42.93%, indicating significant challenges and hurdles for students attempting this module. This highlights a clear area requiring immediate curriculum review or supplementary student support.

****10. Five Recommendations for Management****

- 1. ****Targeted Intervention for At-Risk Students**** Implement a proactive outreach program for students identified as at risk, focusing on personalized support and tailored learning resources.
- 2. ****Course Content Review**** Conduct a thorough review of all courses to assess their difficulty level and instructional design. Consider incorporating supplemental materials and remediation strategies.
- 3. ****Improve Engagement Strategies**** Explore strategies to increase student engagement, such as interactive learning activities, peer support groups, and formative assessments.
- 4. ****Strengthen Cultural Relevance**** Tailor course content and teaching methods to better align with the cultural context of Saudi Arabia and other countries with similar demographics.
- 5. ****Data-Driven Assessment**** Implement a robust system for tracking student performance and identifying areas where interventions are needed.

****4. Key Insights & Patterns****

- * A significant portion of the At Risk cohort is comprised of students under 20, indicating a potential need for early intervention and support for this demographic.**
- * The course C030 consistently demonstrates high completion rates, suggesting a quality curriculum and effective teaching.**
- * The correlation between quiz attempts and failure rates highlights the importance of student engagement and comprehension.**
- * The course C039's high failure rate suggests a need for a more rigorous approach to assessment and instruction.**

****5. Strategic Recommendations****

- 1. **Develop a Targeted Retention Program:** Create a program specifically designed to address the needs of the At-Risk cohort, focusing on personalized support and remediation.**
- 2. **Curriculum Enhancement:** Invest in curriculum development and instructional design to address identified gaps in student understanding.**
- 3. **Increase Student Engagement:** Implement interactive learning activities and formative assessment strategies to boost student participation.**
- 4. **Cultural Sensitivity Training:** Provide training to instructors on culturally relevant pedagogy.**
- 5. **Pilot New Strategies:** Test new engagement strategies on a small scale before widespread implementation.**

****6. Conclusion****

This analysis provides a foundational understanding of student performance and identifies key areas for improvement. By implementing the recommendations outlined above, the institution can proactively address student challenges, enhance learning outcomes, and ultimately improve retention rates. Further investigation into specific

course content and student demographics will be crucial to refining these strategies and maximizing impact.